

SOLAR & WIND DEPLOYMENT INTELLIGENCE PLATFORM

Project Documentation

**AI-powered renewable energy site intelligence, forecasting and deployment
planning**

Prepared from the supplied project specification and architecture reference

Python • FastAPI • React.js • PostgreSQL/PostGIS • GIS • Machine Learning • Docker

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Document Control

Item	Details
Project	Solar & Wind Deployment Intelligence Platform
Document type	Technical Project Documentation
Purpose	Renewable-energy site intelligence and deployment planning
Architecture	Layered web platform with API, data, AI/ML, GIS and dashboard components
Reference basis	Supplied 17-page project specification
Version	1.0

This document describes the platform purpose, architecture, functional modules, data and analytics workflow, technology stack, security approach, testing strategy, deployment model, performance metrics, implementation roadmap and future scope. Module names and scoring weights follow the supplied project specification.

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1. Executive Summary

The Solar & Wind Deployment Intelligence Platform is an AI-powered decision-support system for renewable-energy deployment. Its objective is to recommend suitable locations for solar and wind projects by combining environmental, geographic, climatic and infrastructure-related factors. The supplied specification describes a platform that brings together geospatial analytics, satellite imagery, weather forecasting, terrain analysis, machine learning and optimization workflows.

The platform is intended for renewable-energy companies, government agencies, utility providers, environmental organizations, infrastructure planners and sustainability consultants. Its outputs include site suitability assessments, renewable-resource predictions, energy-generation forecasts, deployment recommendations, feasibility information and decision dashboards.

Expected outcomes

- AI-powered renewable-energy intelligence platform.
- Secure authentication and role-based access.
- GIS and environmental data analysis workflows.
- Solar and wind potential prediction.
- Site suitability and deployment optimization.
- Energy forecasting and investment analytics.
- Role-specific dashboards, reports and alerts.
- Docker-based deployment and operational monitoring.

2. Problem Statement and Objectives

Renewable-energy projects require decisions about resource availability, terrain, infrastructure access, environmental constraints and economic feasibility. These factors originate from different datasets and can be difficult to evaluate consistently. The proposed platform addresses this by providing a unified workflow for collecting data, processing spatial information, calculating renewable-resource indicators, scoring candidate sites and supporting deployment planning.

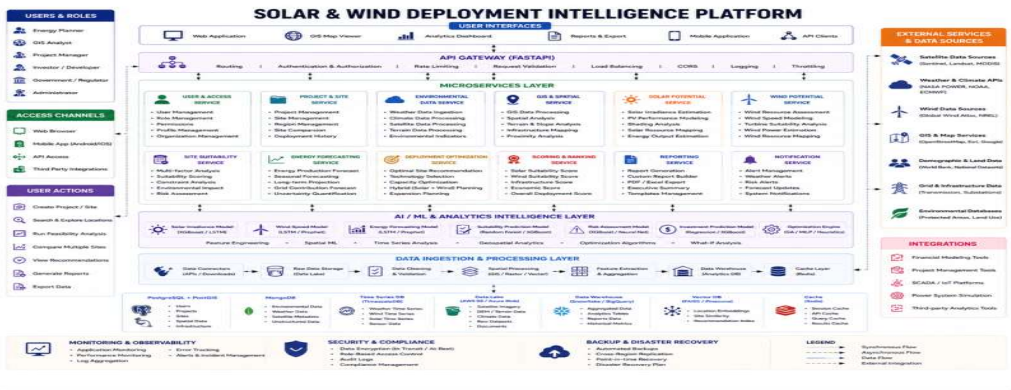
Core objectives

- Analyze solar irradiance, wind speed, temperature, rainfall, cloud cover and related environmental variables.
- Process roads, transmission lines, substations, urban areas, protected zones, water bodies and agricultural land.
- Evaluate elevation, land slope and other terrain characteristics.
- Estimate solar and wind potential through prediction and forecasting workflows.
- Rank candidate sites using a weighted multi-factor suitability model.
- Support technology selection, capacity planning, hybrid recommendations and expansion planning.
- Provide dashboards and PDF/Excel reporting for different user roles.

3. System Architecture

The architecture diagram in the supplied reference (page 2) shows user interfaces connected through an API gateway to application/microservice layers, AI/analytics services and data-ingestion pipelines. External satellite, weather, GIS and environmental sources feed the platform. The diagram also identifies monitoring, security/compliance, backup and disaster-recovery components.

3. Architecture Diagram:



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- User registration and login
- JWT authentication
- OAuth2 login
- Role-based access control
- User profile management

Roles

- Renewable Energy Planner

Figure 1. Architecture diagram from the supplied project specification, page 2.

Major layers

- **User interface:** web application, GIS/map view, analytics dashboards, reports/export and client access.
- **API/application:** authentication, project/site management, environmental data, site intelligence, forecasting, reports and notifications.
- **AI & analytics:** solar/wind prediction, geospatial analysis, suitability scoring, forecasting, optimization and investment analytics.
- **Data ingestion:** weather, satellite, terrain and geographic data collection and transformation.
- **Data layer:** PostgreSQL + PostGIS as the primary database and MongoDB as a secondary database in the specification.
- **Operations:** monitoring, security/compliance, backup and disaster recovery.

4. Functional Modules

The specification defines fourteen major modules forming an end-to-end workflow from authentication and site registration to prediction, optimization, reporting and deployment.

4.1 User Authentication & Role-Based Access

- Registration and login
- JWT authentication
- OAuth2 login
- Role-based access control
- User profile management
- Roles: Renewable Energy Planner, GIS Analyst, Project Manager, Administrator

4.2 Project & Site Management

- Project creation
- Site registration
- Region management
- Site comparison
- Deployment history
- Site information: project ID, coordinates, region, land area, elevation, infrastructure and land ownership

4.3 Environmental Data Collection Engine

- Weather collection
- Satellite image processing
- Terrain analysis
- Climate integration
- Geographic information analysis

4.4 Geographic Intelligence Engine

- GIS processing
- Terrain mapping
- Accessibility analysis
- Infrastructure proximity
- Land suitability assessment

4.5 Solar Potential Prediction Engine

- Solar energy estimation
- Panel-efficiency prediction
- Seasonal forecasting
- Shading analysis
- Solar resource mapping

4.6 Wind Potential Prediction Engine

- Wind resource assessment

- Turbine suitability
- Wind-power estimation
- Seasonal forecasting
- Wind resource mapping

4.7 Site Suitability Intelligence Engine

- Site ranking
- Multi-factor suitability
- Deployment feasibility
- Environmental impact evaluation
- Investment prioritization

4.8 Energy Forecasting Engine

- Energy production forecasting
- Seasonal generation prediction
- Long-term energy estimation
- Grid contribution forecasting
- Revenue prediction

4.9 Deployment Optimization Engine

- Optimal location recommendation
- Technology selection
- Capacity planning
- Hybrid solar-wind recommendations
- Expansion planning

4.10 Site Scoring Engine

- Solar suitability score
- Wind suitability score
- Infrastructure score
- Investment score
- Overall deployment score

4.11 Dashboard & Analytics

- Energy Planner, GIS Analyst, Project Manager and Admin dashboards

4.12 Notification & Alert System

- Weather alerts
- Suitability updates
- Environmental risk alerts
- Forecast updates
- Project notifications

4.13 Reports & Export System

- Site assessment

- Solar potential
- Wind potential
- Feasibility
- Investment reports
- PDF and Excel export

4.14 Integration, Testing & Deployment

- Frontend/backend integration
- API validation
- End-to-end testing
- Security testing
- Performance optimization
- Dockerization
- Production deployment
- Monitoring and documentation

5. Data Sources and GIS Processing

The project specification recommends the following data sources and APIs. Their stated purposes are reproduced here; actual availability, licensing, refresh rate and credentials should be verified for production use.

Source	Purpose
NASA POWER	Solar irradiance analysis and climate data collection
Global Wind Atlas	Wind resource assessment and wind-speed prediction
NASA SRTM Elevation	Terrain analysis and elevation mapping
OpenStreetMap (OSM)	Road networks and infrastructure mapping
Copernicus Sentinel	Land-cover analysis and environmental monitoring
NASA POWER API	Environmental/solar data integration
OpenWeather API	Weather information
Copernicus Sentinel Hub	Satellite data access
OpenStreetMap APIs	Geographic and infrastructure information

Environmental factors

- Solar irradiance
- Wind speed and direction
- Temperature
- Rainfall
- Cloud cover
- Elevation and land slope
- Vegetation index

Geographic features

- Roads, transmission lines and substations
- Urban areas
- Protected zones and water bodies
- Agricultural land

6. AI/ML and Intelligence Engines

The reference specification lists XGBoost, Random Forest, LightGBM, TensorFlow, PyTorch and scikit-learn as possible AI/ML technologies, with Pandas and NumPy for data analytics. It does not define one mandatory model for each task.

Solar intelligence

- Estimate solar-energy potential from environmental and geographic variables.
- Generate annual irradiance, peak sun hours, expected energy output, capacity factor and performance ratio.
- Support seasonal forecasting, shading analysis and solar resource mapping.

Wind intelligence

- Assess wind resources and turbine suitability.
- Estimate wind power and expected annual energy production.
- Generate average wind speed, wind power density, turbulence intensity and capacity-factor metrics.

Forecasting and optimization

- Forecast seasonal and long-term energy production and related revenue/grid metrics.
- Recommend location, technology, capacity, hybrid configuration and expansion options.
- The exact trained model architecture, features, weights and validation results are implementation-specific and are not defined in the supplied reference.

7. Site Suitability Scoring

The supplied specification defines the following weighted model:

Deployment Suitability Score = Renewable Resource Availability (35%) + Geographic Suitability (25%) + Infrastructure Accessibility (15%) + Environmental Impact (15%) + Economic Feasibility (10%)

Factor	Weight
Renewable Resource Availability	35%
Geographic Suitability	25%
Infrastructure Accessibility	15%
Environmental Impact	15%
Economic Feasibility	10%
Total	100%

Suitability categories

- Excellent
- Highly Suitable
- Moderately Suitable
- Low Suitability
- Unsuitable

For a production implementation, each component should have a documented normalization method and validated thresholds for the five categories.

8. Dashboards, Reports and Notifications

The platform is organized around role-specific information needs.

Energy Planner Dashboard

- Recommended deployment sites
- Energy generation forecasts
- Site suitability scores
- Investment recommendations

GIS Analyst Dashboard

- GIS visualization
- Environmental analytics
- Terrain maps
- Site comparison reports

Project Manager Dashboard

- Project progress
- Feasibility reports
- Cost-benefit analysis
- Deployment timelines

Administrator Dashboard

- User management
- Platform analytics
- Data-source management
- System monitoring

Alerts and exports

- Weather and environmental-risk alerts
- Suitability and forecast updates
- Project notifications
- Site, solar, wind, feasibility and investment reports
- PDF and Excel export

9. Technology Stack

The supplied specification lists the following technologies. The final implementation should use the subset actually configured in the project codebase.

Area	Technologies listed
Backend	Python, FastAPI
Frontend	JavaScript, React.js, Next.js, Tailwind CSS
Databases	PostgreSQL + PostGIS; MongoDB
Authentication	JWT Authentication; OAuth2
Machine Learning	XGBoost, Random Forest, LightGBM, TensorFlow, PyTorch, scikit-learn
Analytics	Pandas, NumPy
GIS	QGIS, GDAL, Rasterio, GeoPandas, Shapely
APIs	NASA POWER, OpenWeather, Copernicus Sentinel Hub, OpenStreetMap
Visualization	Plotly, Leaflet.js, Mapbox, Chart.js
Cloud/DevOps	Docker, AWS/Azure, Docker Compose, GitHub Actions
Development/Testing	VS Code, Git & GitHub, Postman

10. Data Model and Core Entities

The specification requires project/site management and identifies PostgreSQL + PostGIS as the primary database. The following are conceptual entities derived from those workflows; exact table names and fields should follow the implemented schema.

User: identity, authentication, role and profile information.

Role: planner, GIS analyst, project manager or administrator access category.

Project: project identifier, metadata, region and deployment history.

Site: candidate location, coordinates, land area, elevation, infrastructure and ownership.

Environmental Observation: weather, climate, solar, wind and related environmental variables.

Geospatial Feature: roads, transmission lines, substations, protected zones, water bodies and land-use features.

Resource Assessment: solar/wind metrics and prediction outputs.

Suitability Assessment: component scores, weighted score and suitability category.

Forecast: energy production, seasonal/long-term estimates and revenue-related outputs.

Recommendation: location, technology, capacity or hybrid configuration.

Report: generated site/resource/feasibility/investment report.

Notification: weather, environmental, forecast, site or project alert.

11. Security and Access Control

Security is a named concern in the reference architecture. The functional requirements include registration/login, JWT authentication, OAuth2 login and role-based access control.

Security controls

- Use strong one-way password hashing; never store plaintext passwords.
- Use secure token handling and expiration.
- Enforce authorization on protected API operations.
- Validate request data and uploaded/exported content.
- Keep secrets and API credentials out of source code.
- Use HTTPS/TLS in production.
- Apply database least privilege and backups.
- Log security events without exposing passwords or tokens.
- Separate administrator operations from ordinary planner/analyst functions.

12. Application Workflow

The reference architecture places an API gateway between interfaces and application services. A typical deployment-planning workflow is:

Step 1. Authenticate the user.

Step 2. Create/select a renewable-energy project.

Step 3. Register candidate sites with geographic metadata.

Step 4. Retrieve or ingest environmental and geographic data.

Step 5. Run GIS processing for terrain, accessibility and infrastructure relationships.

Step 6. Run solar and/or wind resource analysis.

Step 7. Calculate suitability components and the weighted deployment score.

Step 8. Run energy forecasting and related analytics.

Step 9. Run optimization for location, technology and capacity recommendations.

Step 10. Present results through dashboards and generate reports.

API documentation note

The supplied specification does not provide a definitive endpoint inventory. Therefore this document does not invent endpoint names, HTTP methods or request/response schemas; those should be documented directly from the implemented FastAPI routes and frontend service calls.

13. Testing and Validation

Testing is explicitly included in the roadmap and evaluation criteria. Validation should cover the complete workflow as well as individual services.

Testing levels

- Unit testing of authentication, scoring, data transformations and analytics functions.
- API testing of validation, authentication/authorization, errors and responses.
- Integration testing of frontend/backend, database, GIS and external-data adapters.
- End-to-end testing from login through recommendation and reporting.
- Security testing of invalid credentials, unauthorized roles, token handling and input validation.
- Performance testing of API response time, dashboard loading, GIS latency and concurrency.

Acceptance evidence

- Users can authenticate through the configured mechanism.
- Protected functions enforce roles.
- Candidate sites can be stored with geographic metadata.
- GIS/environmental processing produces documented outputs.
- Suitability scores reproduce the documented weights.
- Dashboards and reports use the same assessment results.
- Containerized services start consistently.

14. Docker Deployment and Operations

Docker and Docker Compose are identified as deployment technologies. Containerization separates services and makes the environment reproducible.

Deployment components

- Database service for persistent project and spatial data.
- Backend/API service for authentication, business logic and analytics orchestration.
- Frontend service for the web interface.
- Optional dedicated ML service when prediction workloads are separated.
- Persistent volumes and environment-based configuration.

Operational practices

- Keep secrets outside source code.
- Persist database data through managed volumes or managed databases.
- Use health checks and structured logs.
- Back up critical data.
- Use CI/CD for repeatable builds and tests.
- For cloud deployment, use TLS, network controls, resource limits and monitoring.

Typical startup flow

Obtain source → configure environment variables → build images → start with Docker Compose → verify service health → open frontend → authenticate → test the project/site workflow.

15. Performance Metrics and Evaluation

The reference defines evaluation categories but does not provide measured results. The following are metrics to evaluate, not claims of achieved performance.

Category	Metric
Solar prediction	Irradiance accuracy; energy-generation estimation accuracy; capacity-factor prediction error
Wind prediction	Wind-speed accuracy; wind-power estimation accuracy; seasonal forecast accuracy
Site selection	Suitability classification accuracy; recommendation precision; infrastructure assessment accuracy
Forecasting	Annual energy prediction accuracy; revenue estimation accuracy; investment recommendation effectiveness
System performance	GIS processing latency; API response time; dashboard loading speed; concurrent geospatial-analysis capacity

Quantitative goals include recommending suitable renewable-energy locations, estimating long-term generation potential, supporting feasibility/investment intelligence, optimizing infrastructure placement while considering environmental impact, and supporting large-scale geospatial analytics with stable responsiveness.

16. Implementation Roadmap

Milestone	Main work	Expected outcome
Weeks 1–2	Objectives, architecture, schema, UI wireframes, environments, authentication, project setup, site management, GIS environmental integration	Core setup, site workflows, GIS management
Weeks 3–4	Environmental engine, GIS processing, solar prediction, wind estimation, resource reports	Environmental intelligence and resource prediction
Weeks 5–6	Suitability, optimization, forecasting, investment recommendations, dashboards	Site intelligence and recommendation workflows
Weeks 7–8	Dashboards, reports, GIS visualization, testing, Docker/cloud deployment, end-to-end demonstration	End-to-end demonstration platform

Evaluation checkpoints

- Week 2: initialization, authentication, site management and environmental integration.
- Week 4: solar prediction, wind prediction and GIS analytics.
- Week 6: site suitability, forecasting and optimization.
- Week 8: deployed frontend/backend, dashboards/reporting and end-to-end demonstration.

17. Limitations and Future Enhancements

The supplied specification is a high-level project definition. It identifies modules, data sources, tools and evaluation categories but does not fully specify trained model artifacts, exact scoring thresholds, API contracts, database DDL, measured benchmarks or cloud infrastructure configuration.

Future enhancements

- Calibrate prediction models with region-specific historical data.
- Add uncertainty estimates to forecasts and recommendations.
- Make scoring weights configurable while retaining an auditable default.
- Expand land-use, environmental-impact and regulatory analysis.
- Add detailed CAPEX/OPEX, tariff, financing and payback models.
- Add explainability showing which factors influence site scores.
- Automate data refresh and data-quality monitoring.
- Scale GIS/ML workloads independently in cloud infrastructure.
- Expand automated testing and continuous deployment.

18. Conclusion

The Solar & Wind Deployment Intelligence Platform provides a structured framework for renewable-energy deployment planning. It connects authentication, project/site management, environmental and geographic intelligence, renewable-resource prediction, suitability scoring, forecasting, optimization, dashboards, reporting and operational controls.

The supplied architecture and module plan establish an end-to-end decision-support workflow in which spatial and environmental evidence is transformed into consistent site assessments and deployment recommendations. The weighted scoring model and evaluation metrics provide a transparent basis for validation.

For production use, the platform should be complemented with verified datasets, documented model-training procedures, explicit API/database contracts, security controls, measured performance results and operational monitoring.

19. References

Primary project reference

Solar & Wind Deployment Intelligence Platform — supplied 17-page project specification and architecture reference. It contains the stated objectives, architecture diagram, modules, recommended datasets, technology stack, performance metrics, milestones and quantitative goals.

Named data and technology sources in the specification

- NASA POWER
- Global Wind Atlas
- NASA SRTM
- OpenStreetMap
- Copernicus Sentinel / Sentinel Hub
- OpenWeather
- PostgreSQL / PostGIS
- FastAPI
- React.js
- Docker / Docker Compose
- QGIS, GDAL, Rasterio, GeoPandas and Shapely

Note: URLs, credentials and version-specific documentation are not reproduced because the supplied reference names these sources but does not provide verified URLs or credentials.